The -Xms flag sets the initial heap memory, and the -Xmx adjusts the maximum heap memory. The unwritten rule is not to let the -Xms and -Xmx settings exceed 50% of the node's total RAM. Apache Lucene running under the hood uses the second half of the memory for its segmentation, caching, and other processes.

As we now know, Elasticsearch is a distributed cluster with a master node controlling the cluster while the rest of the nodes perform their individual jobs. A great deal of thought has gone into designing and developing the master node and its features, and we discuss it in its own section next.

14.6 Cluster masters

Every node in a cluster can have multiple roles assigned: `master`, `data`, `ingest`, `ml` (machine learning), and others. Assigning the master role indicates that this node is a *master-eligible* node. Before we discuss master eligibility, let's look at the importance of a master node.

14.6.1 Master nodes

The master node is responsible for cluster-wide operations such as allocating shards to nodes, index management, and other lightweight operations. The master node is a critical component responsible for keeping the cluster healthy. It strives to keep the state of the cluster and node community intact. Only one master node exists for a cluster, and its sole job is to worry about cluster operations—nothing more, nothing less.

Master-eligible nodes are nodes that are tagged with the `master` role. Assigning a `master` role to a node doesn't mean the node becomes the cluster master, but it is one step closer to becoming one, should the elected master crash. The other master-eligible nodes are also in line to become a master, given an opportunity, so they are also one step closer to becoming the master.

What's the use of a master-eligible node, you may ask? Every master-eligible node exercises its vote to select a cluster's master. Behind the scenes, when we boot up nodes for the first time to form a cluster or when the master dies, one of the first steps is to elect a master. We examine elections for the master cluster in the following section.

14.6.2 Master elections

The cluster master is chosen democratically by election! An election is held to choose a master when the cluster is formed for the first time or when the current master dies. If the master crashes for whatever reason, the master-eligible nodes call for an election. Members cast their vote to elect a new master. Once elected, the master node takes over the duties of cluster management.

Not all days are happy days—circumstances beyond its control that can knock off a master node. Therefore, the master-eligible nodes communicate constantly with the master node to be sure it is alive and notify the master of their status. When the master node is gone, the immediate job of the master-eligible nodes is to call for elections to elect a new master.